



DECISION TREE

Lecture 02 | Week 01

by

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DATA

ID	Age	Has_Job	Own_House	Credit_Rating	Class
01	young	false	false	fair	No
02	young	false	false	good	No
03	young	true	false	good	Yes
04	young	true	true	fair	Yes
05	Young	false	false	fair	No
06	Middle	false	false	fair	No
07	Middle	false	false	good	No
08	Middle	true	true	good	Yes
09	Middle	false	true	excellent	Yes
10	Middle	false	true	excellent	Yes
11	Old	false	true	excellent	Yes
12	Old	false	true	good	Yes
13	Old	true	false	good	Yes
14	Old	true	false	excellent	Yes
15	Old	false	false	fair	No

CALCULATING ENTROPY OF THE DATASET

ID	Age	Has_Job	Own_House	Credit_Rating	Class
03	young	true	false	good	Yes
04	young	true	true	fair	Yes
08	Middle	true	true	good	Yes
09	Middle	false	true	excellent	Yes
10	Middle	false	true	excellent	Yes
11	Old	false	true	excellent	Yes
12	Old	false	true	good	Yes
13	Old	true	false	good	Yes
14	Old	true	false	excellent	Yes

ID	Age	Has_Job	Own_House	Credit_Rating	Class
01	young	false	false	fair	No
02	young	false	false	good	No
05	Young	false	false	fair	No
06	Middle	false	false	fair	No
07	Middle	false	false	good	No
15	Old	false	false	fair	No

$$Entropy_{Dataset} = - \left[\frac{9}{15} \log_2 \frac{9}{15} \right] - \left[\frac{6}{15} \log_2 \frac{6}{15} \right] = \mathbf{0.971}$$

Calculating Entropy for Age

ID	Age	Class
01	young	No
02	young	No
03	young	Yes
04	young	Yes
05	young	No

$$Entropy_{Age = young} = -\left(\frac{2}{5}\log_2\frac{2}{5} + \frac{3}{5}\log_2\frac{3}{5}\right) = 0.971$$

ID	Age	Class
06	middle	No
07	middle	No
08	middle	Yes
09	middle	Yes
10	middle	Yes

$$Entropy_{Age = middle} = -\left(\frac{3}{5}\log_2\frac{3}{5} + \frac{2}{5}\log_2\frac{2}{5}\right) = 0.971$$

ID	Age	Class
11	old	Yes
12	old	Yes
13	old	Yes
14	old	Yes
15	old	No

$$Entropy_{Age = old} = -\left(\frac{4}{5}\log_2\frac{4}{5} + \frac{1}{5}\log_2\frac{1}{5}\right) = 0.722$$

Entropy Split | Age

$$Entropy Split_{Age} = \left[\frac{5}{15} \times 0.971 \right] + \left[\frac{5}{15} \times 0.971 \right] + \left[\frac{5}{15} \times 0.722 \right] = 0.888$$

Information Gain | Age

$$I.G_{Age} = Entropy_{Dataset} - Entropy Split_{Age} = 0.971 - 0.888 = \mathbf{0.083}$$

Calculating Entropy for Has_Job

ID	Has_Job	Class
03	true	Yes
04	true	Yes
08	true	Yes
13	true	Yes
14	true	Yes

$$Entropy_{Has_Job = true} = -\left(\frac{5}{5}\log_2\frac{5}{5} + \frac{0}{5}\log_2\frac{0}{5}\right) = 0$$

ID	Has_Job	Class
01	false	No
02	false	No
05	false	No
06	false	No
07	false	No
09	false	Yes
10	false	Yes
11	false	Yes
12	false	Yes
15	false	No

$$Entropy_{Has_Job = false} = -\left(\frac{4}{10}\log_2\frac{4}{10} + \frac{6}{10}\log_2\frac{6}{10}\right) = 0.970$$

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Entropy Split | Has_Job

$$Entropy Split_{Has_Job} = \left[\frac{5}{15} \times 0 \right] + \left[\frac{10}{15} \times 0.970 \right] = 0.646$$

Information Gain | Has_Job

$$I.G_{Has_Job} = Entropy_{Dataset} - Entropy Split_{Has_Job} = 0.971 - 0.646 = \mathbf{0.325}$$

Calculating Entropy for Own_House

ID	Own_House	Class
04	true	Yes
08	true	Yes
09	true	Yes
10	true	Yes
11	true	Yes
12	true	Yes

$$Entropy_{Own_House = true} = -\left(\frac{6}{6}\log_2\frac{6}{6} + \frac{0}{6}\log_2\frac{0}{6}\right) = 0$$

ID	Own_House	Class
01	false	No
02	false	No
03	false	Yes
05	false	No
06	false	No
07	false	No
13	false	Yes
14	false	Yes
15	false	No

$$Entropy_{Own_House = false} = -\left(\frac{3}{9}\log_2\frac{3}{9} + \frac{6}{9}\log_2\frac{6}{9}\right) = 0.918$$

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Entropy Split | Own_House

$$Entropy Split_{Own_House} = \left[\frac{6}{15} \times 0 \right] + \left[\frac{9}{15} \times 0.918 \right] = 0.550$$

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Information Gain | Own_House

$$I.G_{Own_House} = Entropy_{Dataset} - Entropy Split_{Own_House} = 0.971 - 0.550 = \mathbf{0.421}$$

Calculating Entropy for Credit_Rating

ID	Credit_Rating	Class
01	fair	No
04	fair	Yes
05	fair	No
06	fair	No
15	fair	No

$$Entropy_{Credit_Rating = fair} = -\left(\frac{1}{5}\log_2\frac{1}{5} + \frac{4}{5}\log_2\frac{4}{5}\right) = 0.721$$

ID	Credit_Rating	Class
02	good	No
03	good	Yes
07	good	No
08	good	Yes
12	good	Yes
13	good	Yes

$$Entropy_{Credit_Rating = good} = -\left(\frac{4}{6}\log_2\frac{4}{6} + \frac{2}{6}\log_2\frac{2}{6}\right) = 0.918$$

ID	Credit_Rating	Class
09	excellent	Yes
10	excellent	Yes
11	excellent	Yes
14	excellent	Yes

$$Entropy_{Credit_Rating = excellent} = -\left(\frac{4}{4}\log_2\frac{4}{4} + \frac{0}{4}\log_2\frac{0}{4}\right) = 0$$

Entropy Split | Credit_Rating

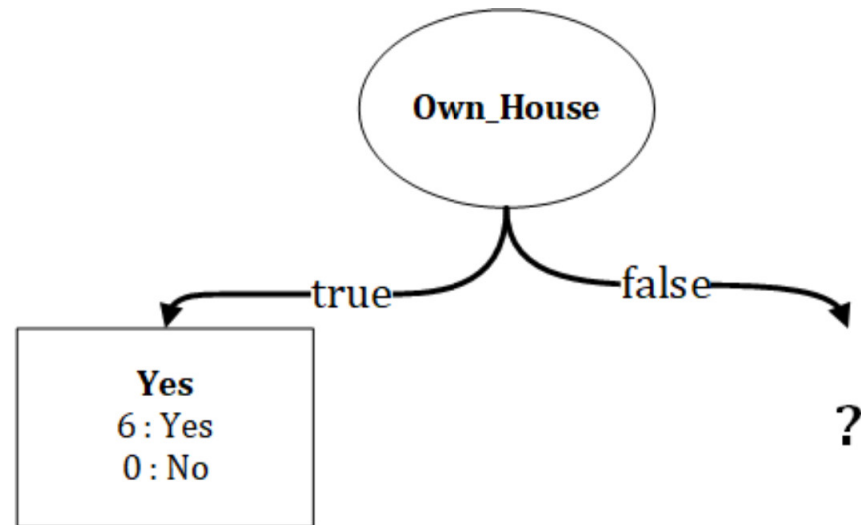
$$Entropy\ Split_{Credit_Rating} = \left[\frac{5}{15} \times 0.721 \right] + \left[\frac{6}{15} \times 0.918 \right] + \left[\frac{4}{15} \times 0 \right] = 0.607$$

Information Gain | Credit_Rating

$$I. G_{Credit_Rating} = Entrop_{Dataset} - Entropy\ Split_{Credit_Rating} = 0.971 - 0.607 = \mathbf{0.364}$$

$$I.G_{\text{Own_House}} > I.G_{\text{Credit_Rating}} > I.G_{\text{Has_Job}} > I.G_{\text{Age}}$$

$$0.421 > 0.364 > 0.325 > 0.083$$



DATA

Calculating Entropy of the Dataset for Own_House → false

ID	Age	Has_Job	Own_House	Credit_Rating	Class
01	young	false	false	fair	No
02	young	false	false	good	No
03	young	true	false	good	Yes
05	young	false	false	fair	No
06	middle	false	false	fair	No
07	middle	false	false	good	No
13	old	true	false	good	Yes
14	old	true	false	excellent	Yes
15	old	false	false	fair	No

$$Entropy_{Dataset \rightarrow Own_House = false} = - \left[\frac{3}{9} \log_2 \frac{3}{9} \right] - \left[\frac{6}{9} \log_2 \frac{6}{9} \right] = 0.918$$

Calculating Entropy for Own_House → false → Age

ID	Age	Class
01	young	No
02	young	No
03	young	Yes
05	young	No

$$Entropy_{Age = young} = -\left(\frac{1}{4}\log_2\frac{1}{4} + \frac{3}{4}\log_2\frac{3}{4}\right) = 0.811$$

ID	Age	Class
06	middle	No
07	middle	No

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$$Entropy_{Age = middle} = -\left(\frac{0}{2}\log_2\frac{0}{2} + \frac{2}{2}\log_2\frac{2}{2}\right) = 0$$

ID	Age	Class
13	old	Yes
14	old	Yes
15	old	No

$$Entropy_{Age = old} = -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) = 0.918$$

Entropy Split | Own_House → false → Age

$$Entropy\ Split_{Own_House \rightarrow false \rightarrow Age} = \left[\frac{4}{9} \times 0.811 \right] + \left[\frac{2}{9} \times 0 \right] + \left[\frac{3}{9} \times 0.918 \right] = 0.666$$

Information Gain | Own_House → false → Age

$$\begin{aligned} I. G_{Own_House \rightarrow false \rightarrow Age} &= Entropy_{Dataset \rightarrow Own_House=false} - Entropy\ Split_{Own_House \rightarrow false \rightarrow Age} \\ &= 0.918 - 0.666 = 0.252 \end{aligned}$$

Calculating Entropy for Own_House → false → Has_Job

ID	Has_Job	Class
03	true	Yes
13	true	Yes
14	true	Yes

$$Entropy_{Has_Job = true} = -\left(\frac{3}{3}\log_2\frac{3}{3} + \frac{0}{3}\log_2\frac{0}{3}\right) = 0$$

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ID	Has_Job	Class
01	false	No
02	false	No
05	false	No
06	false	No
07	false	No
15	false	No

$$Entropy_{Has_Job = false} = -\left(\frac{0}{6}\log_2\frac{0}{6} + \frac{6}{6}\log_2\frac{6}{6}\right) = 0$$

Entropy Split | Own_House → false → Has_Job

$$Entropy\ Split_{Own_House \rightarrow false \rightarrow Has_Job} = \left[\frac{3}{9} \times 0 \right] + \left[\frac{6}{9} \times 0 \right] = 0$$

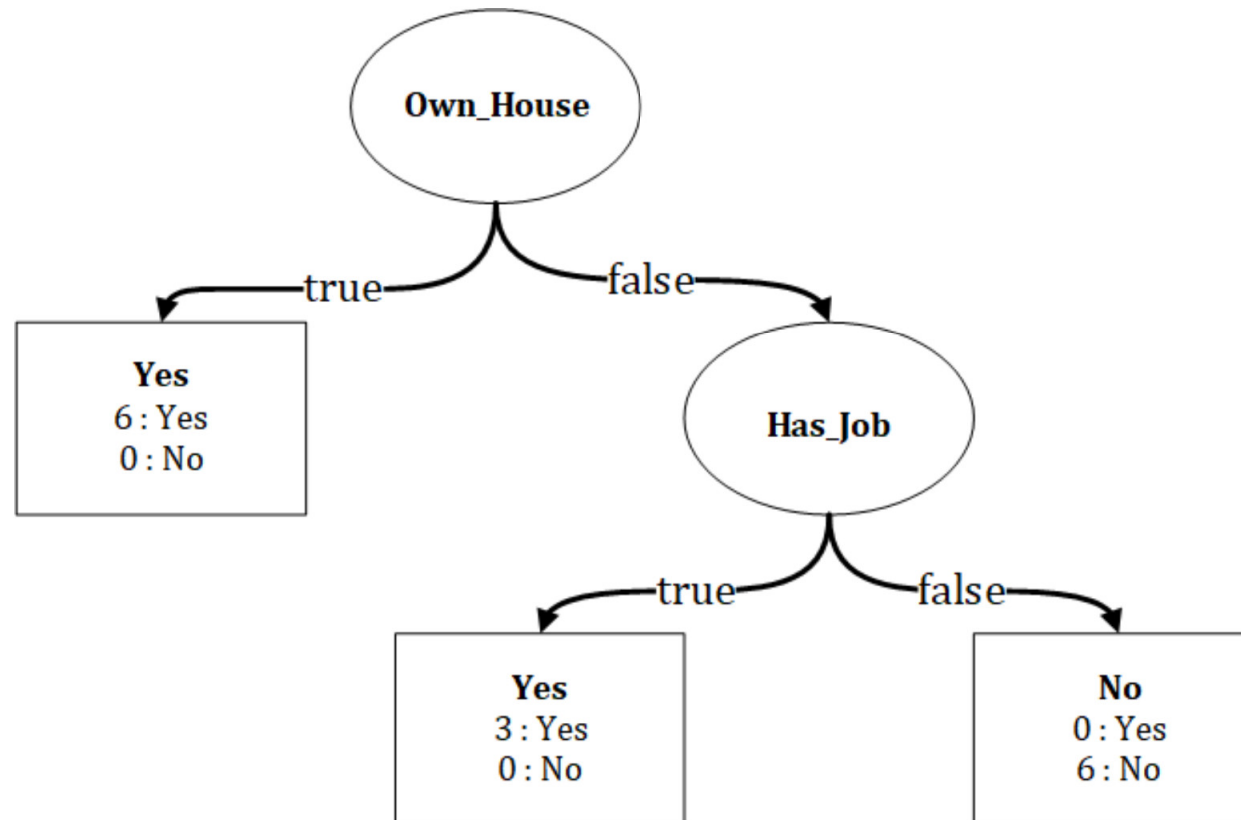
Information Gain | Own_House → false → Has_Job

$$\begin{aligned} I. G_{Own_House \rightarrow false \rightarrow Has_Job} &= Entropy_{Dataset \rightarrow Own_House=false} - Entropy\ Split_{Own_House \rightarrow false \rightarrow Has_Job} \\ &= 0.918 - 0 = \mathbf{0.918} \end{aligned}$$

As it can be seen that:

$$Entropy_{Has_Job = true} = Entropy_{Has_Job = false} = 0$$

Therefore, there is no need to calculate the Entropies of remaining column(s).



THANK YOU!